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## Complex Training: An Update

Kamran Ali\*, Ejaz Hussain M, Shalini Verma, Irshad Ahmad, Deepika Singla and Prakash Jha

### Abstract

The term 'complex training' refers to a training mode which combines one set of strength training with a comparable set of plyometric exercises in same training session and is believed to enhance the quality of the plyometric training stimulus. This idea of combining biomechanically similar exercises is proposed to be a favourable strategy for improving rate of force development and dynamic power by better neuromuscular control. High-intensity resistance training creates an optimal training state for the subsequent plyometric bout by means of neuromuscular, hormonal, metabolic, myogenic, and psychomotor factors that allow it to serve as a strategy for continued neural adaptations. This review aims to present the current body of knowledge regarding the effects of complex training, discussing in brief, its mechanism, the various training variables that might act to confound its efficacy and finally, its comparison with other popular modes of training. In conclusion, complex training appears to be a safe and effective mode yielding gains of both strength and power training in the same session, however, further research is warranted to better understand the operating physiological mechanisms and to derive more concrete results regarding the most optimal training variables.

### Keywords

Complex training; Athletes; Plyometric exercises

### Introduction

The term 'complex training', credited to Russian sport scientist, Yuri Verkhoshansky et al. [1] refers to a training mode which combines one set of strength training with a comparable set of plyometric exercises in same training session and is believed to enhance the quality of the plyometric training stimulus [2-4]. This idea of 2 biomechanically similar exercises performed as a complex is referred to as a complex pair [5] and is proposed to be a favourable strategy for improving rate of force development and dynamic power by better neuromuscular control [6,7].

Complex training is a popular training modality widely used in practical settings and has been examined by a number of reviews [6-9]. Although, most sports scientists and coaches agree that both resistance training (RT) and plyometric training (PT) must be incorporated into athletes' regimens to develop higher muscular power [5], the evidence regarding the efficacy of complex training as an appropriate training modality remains equivocal. Previous literature investigating complex training have shown improvements in athletic performance [10-12], while the opposite has also been reported [13,14]. A possible

explanation for these contradictory findings could be the role of variables, such as the magnitude and mode of the preload exercise, the rest interval between the resistance and plyometric components of complex training. In addition, the gender, training status, training age, strength levels of the participants may also influence the benefits of complex training [7,15].

This review aims to present the current body of knowledge regarding the effects of complex training, discussing in brief, its mechanism, the various training variables that might act to confound its efficacy and finally, its comparison with other popular modes of training.

### Mechanism

The fitness-fatigue theory [16] presents a framework for explaining the mechanism of potentiation associated with complex training. Fitness is the term used to denote the positive response and adaptations following exposure to a training stimulus whereas, fatigue is a term that describes an inability to generate force or sustain further exercise at the required level [17]. Exposure to a training stimulus will result in both a fitness and fatigue response [16], where the former is the desired response, while fatigue is the side effect. Although both fitness and fatigue present some similarities in that they origin from the same source, co-exist [18], rise sharply following training stimulus and resolve thereafter, they exert opposite effects on potentiation and performance. The interplay of these two training responses, determines an athlete's readiness for performance and potentiation which can be modified with strategies that exploit the fitness response and reduce fatigue [16]. Complex training is proposed to be one of these strategies that can help in achieving this preparedness for better performance.

Although the exact physiological rationale behind CT remains inconclusive, CT's ability to heighten the power of the lighter exercises is said thought to result from post activation potentiation (PAP), which is defined as "the phenomenon by which acute muscle force output is enhanced as a result of contractile history and is the premise on which "complex training" is based" [15]. Post activation potentiation (PAP) is the speculated physiological mechanism for complex training [6] and various mechanisms have been put forth to explain its workings. Firstly, the augmentation in plyometric performance after a bout of contractile activity may be attributed to a rise in neural excitability [12]. Docherty et al. [19] also recognise improved motor neuron excitability as the reason for post-activation potentiation. This superior neuronal effect is a consequence of: improved motor unit recruitment, greater motor unit synchronization, amplified central input to the motor neuron, and decreased presynaptic inhibition. On the other hand, phosphorylation of the myosin light chain is another rationale for post-activation potentiation [20]. The quantity of  $Ca^{2+}$  entering the sarcoplasmic reticulum is increased by heavy exercising, which causes the myosin light chain kinase to produces ATP for utilisation by the actin-myosin complex, thereby, increasing the rate of cross bridging between actin and myosin. This phosphorylation makes myofilaments more sensitive to  $Ca^{2+}$ . Power production is improved owing to more ATP production as the level of cellular levels of  $Ca^{2+}$  increase [7]. This theory is supported by Gourgoulis, et al. [21] who adds that complex training might also cause more neurotransmitters

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to be released in afferent nerves. A study by Fees [22], elucidating post-activation potentiation proposed that stimulation of an agonist muscle results in reciprocal inhibition around a joint. The Golgi-tendon organ and Renshaw cell that limit the maximal motor unit activation are allegedly inhibited in reaction to the application of a heavy stimulus to muscles thus allowing recruitment of more motor units resulting in increased power output. Baker [2] suggests this to be another plausible explanation for the operating of post-activation potentiation.

Docherty et al. [7] noted, however, that it is possible that PAP is the result of interactions between both the neural and muscular mechanisms. In addition to neuromuscular, Ebben and Watts [8] listed hormonal, metabolic, myogenic, and psychomotor effects as factors that may contribute to post-activation potentiation, which warrants further explanation.

## Training Variables

### Optimal load

A crucial determinant of the efficiency of a complex training protocol is the intensity of the preload element. Previous researches present uncertainty regarding the optimal load that needs to be lifted in the resistance exercise bout to maximally utilise the PAP. Jeffreys et al. [9] suggests the association of PAP with Type II muscle fibres, requires the preceding exercise bout to stimulate an adequate number of Type II fibres, through either high resistance or high velocity. It is for this reason that, most complex training studies utilise a 5RM squatting protocol. This approach has yielded mixed results wherein, some researchers found this resistive load to have a significant effect on successive plyometric exercise [11, 23], while in others, it produced non-significant changes in the plyometric performance [14, 24].

There is limited research into loads outside of this 5RM range. Baker et al. [2] examined the effect of lifting six repetitions at 65% of 1RM and showed an improvement in performance in a subsequent plyometric exercise. More recently, Mohamed et al. [25] reported enhanced performance in young gymnasts following a resistance exercise of load intensity between 50-60%. On the contrary, Hanson et al. [26] studying the effects of squats at 40% 1RM and 80% 1RM on jumping performance, found no change proposing that these loads were inadequate to provide a PAP effect. Comyns et al. [10] addressed the optimal load issue by investigating the effect of a 65% 1RM, 80% 1RM and 93% 1RM resistance exercise on jump performance. The variables assessed included flight time, ground contact time, peak ground reaction force, and reactive strength index and leg stiffness. Results showed that while all resistive loads significantly reduced flight time, load at 93% intensity also caused a significant improvement in ground contact time and leg stiffness. This might be beneficial from a training perspective where heavy lifting will cause jumps to be performed with a stiffer leg spring action, which in turn may assist performance. Although results from previous findings studying optimal load are conflicting, it would appear that PAP is maximised when heavy loads are utilised.

### ICRI

Rest period is a key component of any training; intra complex rest interval (ICRI) is the time between the resistance and plyometric components [24], which is a major determinant of the efficacy of complex training. The current findings on the optimal ICRI are somewhat ambiguous. Previous studies have used a rest interval has

ranging from 10 seconds [24] up to 20 minutes<sup>13</sup> and indicated that 3 to 4 minutes may be optimum [11,12,23].

Some studies suggest that minimal rest of 10 and 15 seconds seems to decrease subsequent power [24,27]. Jensen & Ebben [24] investigated the effect of squats performed at 5 repetition maximum (5RM) on countermovement jumps (CMJ) that were performed 10 seconds, and 1, 2, 3 and 4 minutes after the squat. Although non-significant, the jump performance was reduced at 10 seconds interval with a trend of improvement from 10 seconds upto 4 minutes. They concluded that 1-4 minutes ICRI did not impair nor improve lower leg power, as assessed using vertical jump. Jones and Lees [13] adopted a similar approach manipulating the length of the rest interval where they investigated the effect on jumps performed immediately, 3, 10 and 20 minutes after 5RM back squats. Results revealed no significant difference, suggesting that complex training did not enhance plyometric performance, it was also noted that no adverse effects occurred.

Comyns et al. [28] used ICRI of 30 seconds, 2, 4 and 6 minutes to find its effects on complex training and showed that 4 minute ICRI is optimal for exploiting Post Activation Potential (PAP), whereas 30 second or 6 minute ICRI decreases subsequent power. More recently, Jeffreys et al. [9] highlighted the findings of Jensen & Ebben [24] proposing that it is likely that optimal PAP will only be evident at a given window of opportunity and outside this performance may be impaired, unchanged or show limited benefits, implying that complex training may be stimulatory or inhibitory depending on the ICRI. In addition, any ergogenic response at the different intervals may be masked by the potentiation window that differs for individuals, as suggested by Docherty et al. [7]. Therefore, individual determination of the intracomplex rest interval may be necessary in the practical setting.

### Individuality

It is crucial to consider individual to individual response when using complex training [7,27,29]. Bevan et al. [30] studied 26 rugby players who performed complex training with an ICRI of 15 seconds, 4, 8,12, 16, 20 and 24 minutes and noticed individualistic response to complex training. 58% reached their greatest peak power output at 8 minutes, 7 subjects at 12 minutes, 3 subjects at 16 minutes and 1 subject at 4 minutes. This emphasises that optimal timeframe for the window of potentiation opportunity is individual so, in a practical setting, the identification of an optimal intracomplex rest interval for group situations is not appropriate and warrants individual determination.

### Gender

Previous studies propose that subjects responded differently to complex training regarding their jump performance with certain ICRI [28] but no differences found between genders regarding optimal ICRI [13,24,28]. Jensen and Ebben [24] assessed the jump performance of 10 women and 11 men following 5RM squats with ICRI of 10 seconds, 1,2,3, and 4 minutes and concluded that the response to ICRI did not vary significantly between the genders. Comyns et al. [28] recruited 9 men and 9 women to investigate the effect on jumps performed 30 seconds, and 2, 4 and 6 minutes post-lifting and assessed flight time and peak ground reaction force (GRF). The improvement window was different for each subject, however, a significant decrease in flight time for men and increase in peak ground reaction force for women was recorded. More recently, Mihalik et al. [31] compared the effects

of complex versus compound training on vertical jumps, performed by 11 men and 20 women and found no significant difference in improvements between genders for either training groups.

## Effects of Training

The effects of complex training may be attributed to several factors, (a) neuromuscular, (b) hormonal, (c) metabolic, (d) myogenic, and (e) psychomotor, that allow it to serve as a strategy for continued neural adaptations, the most powerful mechanisms being neuromuscular. High-intensity resistance training increases motorneuron excitability and reflex potentiation, as the associated fatigue may force more motor units to be recruited during the plyometric phase, thereby creating an optimal training state for the subsequent plyometric bout [6,32,33].

## Physical performance

Numerous studies on complex training have examined its role in the enhancement of measures of power and strength. Athletic performance has been assessed with respect to depth jumps [24], counter movement jumps [13,24,28], hockey sprints [34] and land sprints [35]. The results however, have been equivocal where some studies reported complex training to be beneficial [34] whereas, others show it was not so effective compared to non-complex training interventions [11,13,24,35]. Ebben et al. [36] assessing upper limb plyometric performance using a medicine ball following the horizontal bench press failed to obtain increases in reaction force, electromyographic activity of pectoralis major and triceps brachii. Nevertheless, the authors propose complex training as a good strategy to perform strength and plyometric training in the same session [36]. Gourgoulis et al. [21] identified an increase of 2.39% in vertical jump performance, when the jump was preceded by a squat exercise, with a higher increase in the jump height in athletes with greater maximum strength values. MacDonald et al. [37] also found significant changes in anthropometry in addition to strength gains. Overall, complex training shows same improvement as other strength and power training, with speculated added benefits, making it a viable method for a more efficient workout [38].

## Hormonal responses

Resistance exercise has been shown to induce significant individual, protocol-dependent changes in levels of salivary testosterone and cortisol, emphasizing that, it is highly likely for individual athletes to respond differently to different resistance regimens during their training cycle. Therefore, monitoring these hormonal fluctuations in response to exercise stimuli could prove to be a precise measure to assess stress and manage maximal adaptation. Moreover, the use of saliva provides a novel methodology for trainers and athlete, to non-invasively assess the levels of hormones instrumental in producing adaptations [39]. Beaven et al. [40] studied testosterone and cortisol responses to various resistance training programs, each performed twice a week for 4 weeks. The strength-power session elicited greatest testosterone response than other three groups i.e. power-power, power-strength and strength-strength suggesting that traditional complex training enhanced anabolic hormonal milieu. These results coincide with earlier studies stating that resistance exercise caused salivary hormonal changes with muscle strength gains [39,40]. Recent research showing the association of salivary testosterone responses with strength and gains [41], suggests that the superior functional gains were related to the improvements in the hormonal environment. These findings have practical implications when designing exercise and training to structure workouts in order to achieve maximal benefits [41].

## Sleep

Sleep and exercise demonstrate a bilateral interaction influencing each other through multiple, complex physiological and psychological pathways [42]. The effects of exercise on sleep are controlled by characteristics of the i) individual, including sex, age, fitness level, type of sleeper and body mass (BMI), and ii) exercise protocol, that constitutes intensity, duration, environment, acute or regular, aerobic or anaerobic, with different variables demonstrating contradictory effects [42]. Moreover, because exercise causes significant long-term changes in body composition, basic metabolic rate, cardiac function, glucose control, mood and immune function [43,44], it is actually difficult to differentiate the direct and indirect pathways involved in the exercise-induced changes to sleep. Previous literature examining the interplay of sleep and exercise have emphasized aerobic and strength training and the direct effect of complex training on sleep quality remains to be investigated.

## Complex Vs Other Training Modes

In addition to the immediate effects and benefits of complex training on athletic performance, it is imperative to compare it with other trainings types over the periods of many weeks. Dodd and Alvar [45] in their study used a randomized crossover design to compare the effects of resistance, plyometric and complex training on sprint times, standing broad jumps, vertical jumps and agility. The study period included three 5 week mesocycles, where, participants trained for each protocol for 4 weeks with 1 week of active rest periods. Results demonstrated that complex training brought about benefits of both types of training and demonstrated superior results to both resistance and plyometrics training groups. Mihalik et al. [31] comparing two anaerobic training types- complex and compound training, found an increment of 9% in vertical jump and 7.5 % in mean power in the compound training group, whereas complex training group increased vertical jump by 5.4% and mean power output by 4.8%. There was however, no significant difference in rate of improvement and power could be significantly improved by both the training types in duration as short as 3 weeks. Alves et al. [46] examined the effects of a 6 week complex and contrast training program on soccer performance, when given in addition to regular soccer training. The obtained results suggested that the complex-contrast training is an adequate training strategy to develop soccer players' muscle power and speed and provided significant improvements over the routine sport-specific skill training. Macdonald et al. [37,47] compared the effects of resistance, plyometric and complex training provided twice weekly for 6 weeks, on lower body strength and anthropometrics. The results revealed that complex training mirrors benefits seen with traditional resistance or plyometric training without any detriments to performance making it a viable training modality. Authors also suggest that to obtain significant hypertrophic gains, the intervention of complex training in recreational athletes needed a longer duration of exposure.

## Conclusion

Previous researches [48,49] suggest that strength training must be integrated with sport-specific skills training to improve athletic performance. Although some studies pointed out complex training as a good method to increase sports skills power [10,50,51] others failed to identify this effect [36]. Therefore, in light of current evidence, this strength training method necessitates further investigations to derive conclusive statements about its effectiveness. Moreover, the literature



regarding manipulation of training variables exploiting PAP, as expressed in the enhancement of athletic performance, is very scarce. Most authors have investigated the acute effects of PAP over multiple sets [15,27], which limits the extrapolation of these results to more chronic adaptations. Chronic adaptations induced by PAP warrant detailed examination, to provide a comprehensive understanding of the exact physiological pathway through which it operates to enhance performance. Future studies shall also explore the complex interplay of various variables such as the training load, volume and rest interval, in order to provide concrete data regarding the most optimal protocol.

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